Pulkit Dubey

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Summary

PhD in Applied Mathematics with expertise in turbulence simulation, machine-learning and optimization. Developed high-performance simulations in Python using MPI, CUDA, and PyTorch for scalable computing over HPC clusters. Currently implementing nonlinear control in climate emulators (NeuralGCM, ACE2) using JAX and PyTorch. Cross-functional communication evidenced by collaborative publications and presentations at major conferences.

Experience

University of California, Santa Cruz — **Santa Cruz, CA**
Postdoctoral Researcher — *Oct 2025 - Present*

- **Optimal control of chaotic systems using differentiable neural emulators**: Developed gradient-based optimization pipelines in PyTorch and JAX, using state-of-the-art ML weather emulators (LUCIE, NeuralGCM, ACE2-ERA5) as differentiable surrogate models for atmospheric control. Implemented gradient-based optimization via automatic differentiation, and gradient-free methods such as Reinforcement learning toward control of long horizon targets in chaotic dynamical systems.
- **ML data infrastructure and emulator training for climate applications**: Built end-to-end pipelines for ERA5 reanalysis data, including regridding and quality validation of multi-variable atmospheric fields at global scale. Trained and fine-tuned an SFNO-based neural weather model on ERA5 dataset.

University of New Hampshire — **Durham, NH**
Doctoral Researcher — *Jan 2020 - Sept 2025*

- **Reduced order modeling of turbulent flows**: Developed scalable Python simulation frameworks using spectral methods to handle high-dimensional turbulent flow data from large-scale physics simulations. Leveraged asymptotic analysis for reduced-order modeling, achieving a 10× performance improvement. Implemented custom data structures for efficient post-processing, management, and visualization of simulation statistics. Presented research at APS-DFD 2024; manuscript in preparation for the Journal of Fluid Mechanics.
- **Accelerating numerical solvers with neural networks**: Designed a C++/CUDA framework that interleaves a high-fidelity CFD solver with a PyTorch neural operator: the solver advances 3–5 timesteps, the network forecasts the next window, then returns control to physics. By capping the NN horizon at the chaos limit, the method halves wall-clock time while preserving accuracy and keeps the entire pipeline differentiable for downstream optimization or control tasks. The work was presented at the SC24 supercomputing conference in Atlanta, GA.
- **Teaching assistant for undergraduate courses:** Delivered classroom lectures and mentored students through undergraduate engineering courses. Developed an automated grading system streamlining faculty workload and accelerating student feedback. Recognized with competitive fellowships for outstanding TAs.

TIFR - Center for Applicable Mathematics — **Bangalore, India**
Intern - Airbus Chair for Mathematics of Complex Systems — *Feb 2016 - June 2016*

- **Development of open-source numerical solver in C:** Developed a finite-volume PDE solver for high speed flows in C for the Navier-Stokes equations. Maintained a Git-based workflow for version control, ensuring code portability and collaboration.

Tata Motors — **Pune, India**
Graduate Engineer Trainee — *May 2015 - Jan 2016*

- **Thermal system analyses in IC Engine**: FEM-based studies of crankshaft deflections under extreme loads in ANSYS and CATIA. Compared multiple vendor designs to optimize cost, safety, and structural integrity. Earned Siemens PLM certification.

Education

University of New Hampshire — **Durham, NH**
PhD, Applied Mathematics — *Sept 2025*

Jawaharlal Nehru Centre for Advanced Scientific Research — **Bangalore, India**
M.S.(Engg.) Engineering Mechanics and Applied Mathematics — *May 2019*

BITS - Pilani — **Pilani, India**
B.E.(Hons.) Mechanical Engineering — *August 2015*

Relevant Projects

CAD modeling of MEMS devices: Design of MEMS vibrational energy harvester and gas sensor in Solidworks and coupled thermo-mechanical analysis in ANSYS.

Effects of size of a neural network on overfitting: Approximating mathematical functions with neural networks of varying sizes, starting from a single neuron up to deeper, multi-layer architectures to study overfitting, accuracy and training time. Optimized the training over a CUDA enabled desktop GPU. Gained hands-on experience with the end-to-end ML workflow including model development, training and evaluation.

Neural Surrogate for chaotic dynamical systems: MATLAB based simulation of Lorentz attractor that trains a neural surrogate; benchmarked data-driven predictions against ground truth from ODE solvers to study accuracy and generalization. Extendable to differentiable physics engines for real-time control.

Recognition

Referee for the prestigious journal, Journal of Fluid Mechanics (2023)
Summer fellowship for outstanding TAs (2023,2024)
Fellow for the prestigious Boulder Summer School in Condensed Matter Physics (2022)
Selected for the Airbus Chair on Mathematics of Complex Systems (2016)

Conferences

APS-DFD 2022, 2023, 2024
Boulder School for Condensed Matter Physics 2022
SC24 (International conference on High Performance Computing)

Relevant Publications

Atif, Mohammad, Pulkit Dubey, et al. (2024). "Fourier neural operators for spatiotemporal dynamics in two-dimensional turbulence". In: *arXiv preprint arXiv:2409.14660*.

Dubey, Pulkit, Rama Govindarajan, and Greg Chini (2025). "Quasilinear simulations of viscosity stratified flows". In: *Under preparation for Journal of Fluid Mechanics*.

Dubey, Pulkit, Anubhab Roy, and Ganesh Subramanian (2022). "Linear stability of a rotating liquid column revisited". In: *Journal of Fluid Mechanics* 933, A55. DOI: `10.1017/jfm.2021.1109`.

Kaushal, Shaurya, Pulkit Dubey, et al. (2015). "Novel design for wideband piezoelectric vibrational energy harvester (P-VEH)". In: *2015 19th International Symposium on VLSI Design and Test*. IEEE, pp. 1–5.